MIL-I-27280A(USAF)
22 December 1960
Superseding
MIL-I-27280 (USAF)
13 November 1959

MILITARY SPECIFICATION

INDICATOR, INDICATED AIRSPEED AVU-10/A

1. SCOPE

1.1 This specification covers one type of integrally lighted, pitot static airspeed indicator, designated AVU-10/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-I-5415	Indicators; Airspeed, Pitot and Static Pressure, General Specification for
MIL-P-7936	Parts And Equipment, Aeronautical, Preparation for Delivery
MIL-L-25467	Lighting, Integral, Instrument, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS28105	Window, Dial - Aircraft Instrument Cover, Glass
MS24367	Lamp - Incandescent, Miniature, Integral Lighting

FSC 6610

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MS24515	Lamp, Sub Miniature
MS25237	Lamp, Incadescent, T-1-3/4 Bulb, Midget Flange Base
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33638	Cases, Instrument, Flange-Mounted Aircraft

Air Force-Navy Aeronautical

AND10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 QUALIFICATION.- The indicator furnished under this specification shall be a product which has been tested and has passed the qualification tests herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 GENERAL SPECIFICATION.- The requirements of MIL-I-5415 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.2.1 FLANGE.- The indicator flange shall be 0.500 ±0.156-inch thick.

3.2.2 COVER GLASS.- A MS28105-17 cover glass shall be furnished. The cover glass shall provide an adequate pressure seal and shall be replaceable by removing the bezel. If used as a part of the lighting system, the cover glass size and configuration may deviate from the requirements of MS28105 to the extent required for lighting design. If the cover glass and bezel form an integral assembly, they shall be replaceable as a unit.

3.2.2.1 The distance between the front surface of the bezel and the outside surface of the cover glass shall not exceed 0.125 inch.

3.2.3 CASE.- The case shall be of nonferrous, low-density material and shall be finished in durable dull black.

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3.3 DESIGN.- The indicator shall conform to figures 1 and 2.

3.4 POINTERS.- A MS33585-3 indicated airspeed pointer shall be provided, except that the shade portion of the pointer shall be finished in dull white, color 37875 of FED-STD-595. A MS33585-3 maximum airspeed pointer shall be provided, except that the shaded portion of the pointer shall be finished as shown on figure 3, and the pointer width shall be 0.150 inch in lieu of 0.093 inch. The nonindicating end of the pointers shall not obscure the drum.

3.4.1 The 2 pointers shall be concentric, with the maximum speed pointer closest to the dial and the indicated airspeed pointer nearest the cover glass.

3.5 DIAL.- The dial and drum shall conform to figure 2. The drum shall make 1 revolution for each 100-knot change in airspeed and shall be so geared to the indicated airspeed pointer that proper relationship is maintained at all speeds.

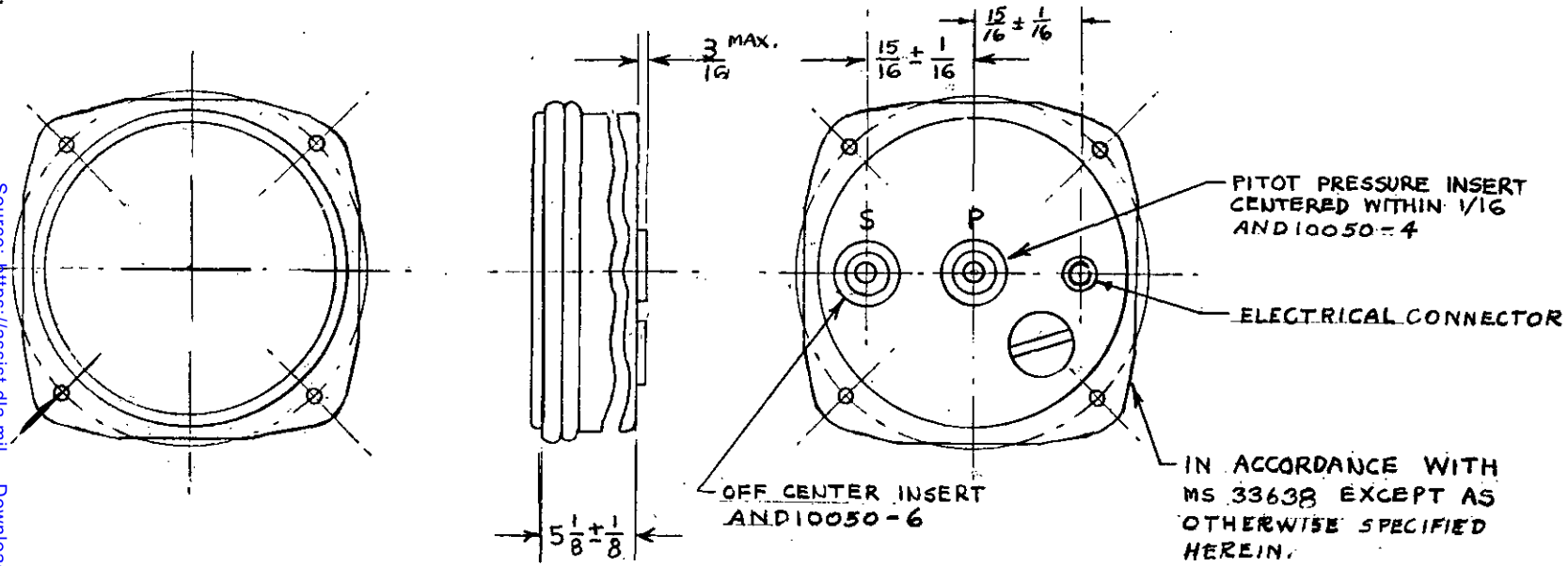
3.5.1 INDEX.- A triangular index conforming to figure 2 shall be provided to show the Mach number to which the instrument is adjusted. The Mach numbers shall be placed on the dial as shown on figure 2 and the mechanism shall be so arranged that the values of the Mach number between the numerals may be set by linear interpolation between the numerals.

3.6 LIMITATION OF POINTER MOVEMENT

3.6.1 AIRSPEED POINTER.- The airspeed pointer movement shall be limited by suitable stops in the mechanism in such a manner that the pointer will not rotate more than 350° of arc clockwise from the NO-PRESSURE position when the indicator is submitted to overpressure tests. The design of the stops shall be subject to the approval of the procuring activity.

3.6.2 MAXIMUM SPEED POINTER.- The maximum speed pointer shall be limited by a stop which shall be adjustable from the front of the indicator by a screw driver slot in the lower left mounting lug. The range of the stop shall be from 300 knots, or less, to 650 knots. The adjustable stop shall be so designed that the stop will in no way affect the indication of the pointer when the altitude and Mach number setting are such as to limit the speed to lower than that set by the adjustable stop.

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FRONT SIDE REAR

DIMENSIONS IN INCHES.

FIGURE 1.- CASE

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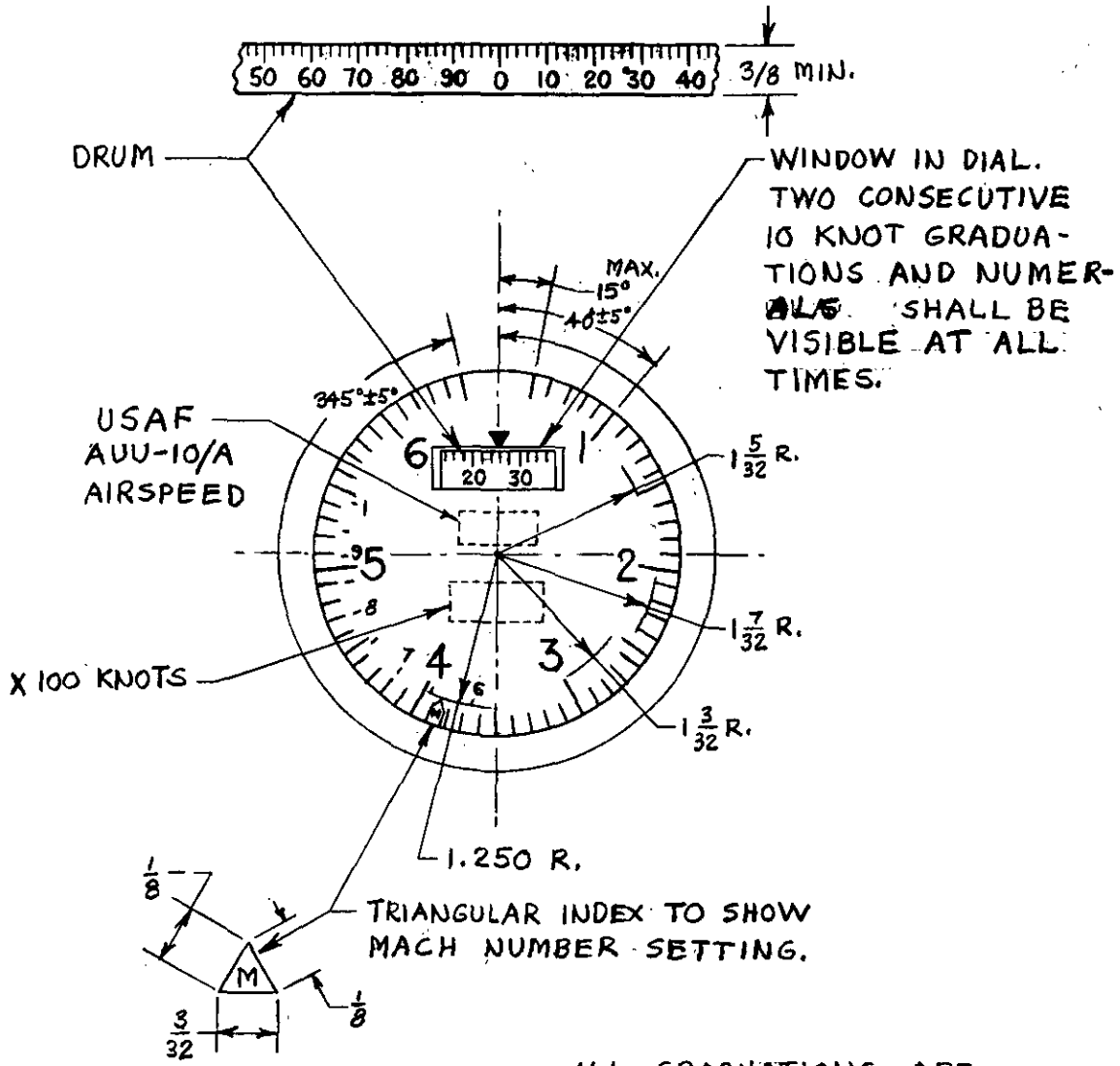


FIGURE 2.- DIAL LAYOUT

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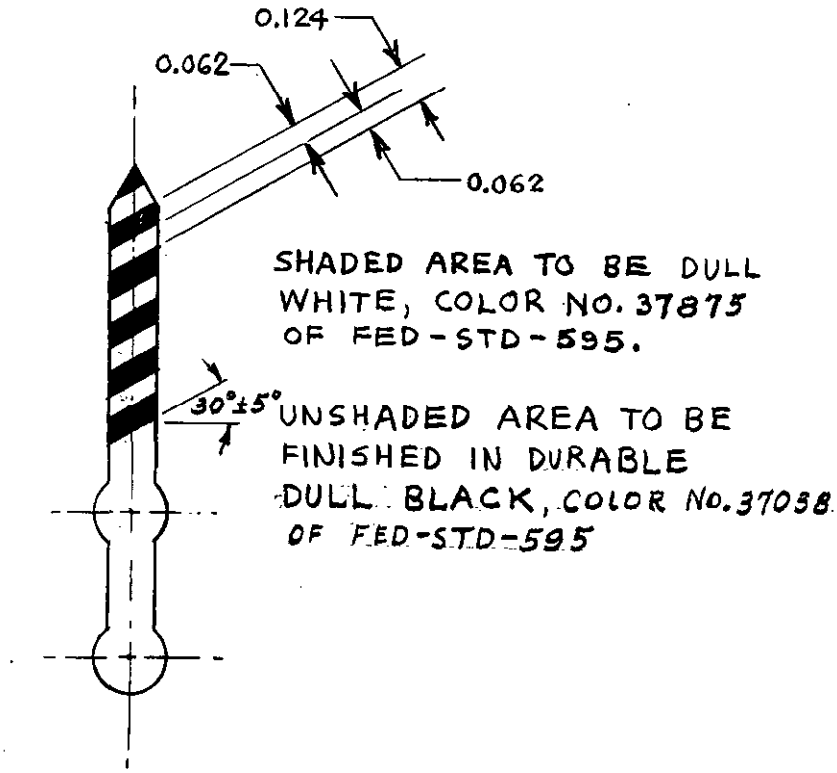


FIGURE 3.- MAXIMUM AIRSPEED POINTER

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3.7 MACH NUMBER ADJUSTMENT.- The Mach number adjustment mechanism shall be accessible by means of a plug located at the rear of the indicator case. The design of the adjustment mechanism shall be subject to the approval of the procuring activity. Adjustment shall be made with the use of standard tools.

3.8 MARKINGS

3.8.1 DURABLE DULL WHITE.- The following markings shall be finished in durable dull white, color 37875 of FED-STD-595. The dimensions of the markings shall be as follows:

	Height or Length Inch ±1/64	Width of Line or Graduation Inch ±0.005
10 Knot graduations	As shown on figure 2	0.016
Triangular index to show drum indication	3/32 on each edge	
Lettering "X100 Knots"	1/8	0.016
Graduations corresponding to numerals and to 50, 150, 250, 350, 450, 550, and 650 knots	As shown on figure 2	0.031
Numerals 1, 2, 3, 4, 5, and 6	5/16 Min.	0.031
Graduations on drum corresponding to numerals	1/8	0.031
2 Knot graduations on drum	3/32	0.016
Numerals on drum	3/16	0.016

3.8.2 DURABLE DULL BLACK.- The lettering USAF, AVU-10/A, and AIRSPEED shall be permanently and legibly marked on the dial. These and all other markings not otherwise specified shall be finished in durable dull black, color 37038 of FED-STD-595. The height at the markings shall be 3/64 ±1/64 inch.

3.8.3 GLOSSY BLACK ENAMEL.- The following markings shall be finished in glossy black, color 17038 of FED-STD-595. The marking dimensions shall be as follows:

	Height or Length Inch ±0.010	Width of Line or graduation Inch ±0.005
Triangular index for Mach No. setting	As shown on figure 2	As shown on figure 2

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	Height or Length Inch ±0.010	Width of Line or graduation Inch ±0.005
Graduations corresponding to Mach numerals	0.062	0.016
Numerals representing Mach number settings	0.093	0.016

3.8.4 VISIBILITY OF DIAL MARKINGS.- All graduations, numerals, pointers, and other markings on the dial shall be readable from any point within the frustum of a cone of which the side makes an angle of not less than 30° with the perpendicular to the dial and the small diameter is the aperture of the dial.

3.8.5 The back of the case, adjacent to the connections, shall be marked as follows:

P	Pressure
S	Static connection
MACH NUMBER ADJUSTMENT	Mach number adjustment cover

3.9 INTEGRAL LIGHTING.- Except as specified herein, the indicator shall be integrally lighted with a red lighting circuit in accordance with MIL-L-25467 (see figure 4 for connections). Although resistors shall not be permitted in the original design of the lighting system, space shall be allocated for incorporation of a trimming resistor in the lighting circuit within the indicator. The magnitude of the trimming resistors shall be specified by the procuring activity in terms of the brightness decrease required. The lighting shall be subject to the approval of the procuring activity.

3.9.1 The integral lighting shall be comparable to and in brightness balance with a prime standard indicator. The prime standard shall be an indicator of the type described herein. The prime standard shall have applied to its lighting circuit the specific voltage which the procuring activity has established as being necessary to assure equality of presentation with respect to the lighting with adjacent but dissimilar instruments. The contractor shall submit the test procedure for determining compliance with this paragraph to the procuring activity for approval.

3.9.2 LAMPS.- MS25237-328, MS24367-5, or MS24515-5 lamps shall be provided. If the MS24367-5 or MS24515-5 lamp is used, the brightness of the lighting system shall be as specified in table I.

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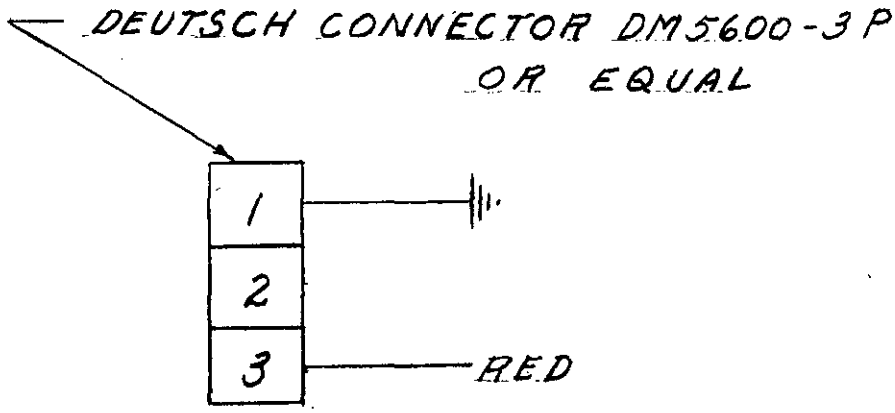


FIGURE 4 - ELECTRICAL CONNECTOR

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TABLE I

Light Intensity and Distribution

Color	Brightness (Foot-Lamberts)
White	1.35 ±.65
Gray	.8 ±.4
Black	.08 ±.05

3.10 WEIGHT.- The weight of the indicator shall not exceed 2.25 pounds.

3.11 IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12 MOUNTING.- No insert or spring nuts shall be furnished.

3.13 WORKMANSHIP.- Workmanship shall be in accordance with high-grade, aircraft instrument manufacturing practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 SUPPLIER RESPONSIBILITY.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 GENERAL SPECIFICATION.- The inspection and testing of indicators shall be in accordance with MIL-I-5415 with the exceptions and additions specified herein.

4.2.1 Unless otherwise specified, for the purpose of this specification the tests specified in MIL-I-5415 shall be identified herein as follows:

MIL-I-5415	MIL-I-27280
Qualification tests	Sampling plan B
Inspection tests	Acceptance tests
Individual tests	Individual tests
Sampling	Sampling plan A

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4.3 TEST CONDITIONS

4.3.1 TEST TOLERANCES.- Test tolerances shall be as specified in tables II and III.

TABLE II

Tests and Test Point Tolerances

Tests and Test Points	Tolerances
Position Error	
Airspeed pointer	3 knots
Maximum speed pointer	3 knots
Friction	
Maximum speed pointer	3 knots
Airspeed pointer	3 knots
Vibration Error	
(Mach number setting at 0.8)	
(Test altitudes of 0 and 20,000 feet)	
(Differential pressures equal to 100 and 400 knots)	
Maximum Vibration (either pointer)	2 knots
Variation from original position (either pointer)	2 knots
Vibration Failure	2 knots
(Test altitude of 20,000 feet)	
(Differential pressure equal to 400 knots)	
(Mach number setting at 0.8)	
Overpressure	2 knots
(Static connection - 6 in. water)	
(Pitot connection - 30 in. Hg)	
(Test altitude of 0 and 20,000 feet)	
(Differential pressure equal to 100 and 400 knots)	
Seasoning	2 knots
Aging	3 knots
Zero position on drum	30 ±10 knots
Damping	
Time required for pointer to travel from full scale deflection to the 60 knots graduation	$1\ ^{+1.0}_{-0.2}$ second

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TABLE III

Speed Tolerances

Speed Knots	Tolerance Knots	Speed Knots	Tolerance Knots
50	4.0	360	4.0
60	4.0*	370	4.0
70	2.0	380	4.0*
80	2.0**	390	4.0
90	2.0	400	5.0**
100	2.0**	410	5.0
110	2.0	420	5.0**
120	2.0*	430	5.0
130	2.0	440	5.0
140	2.0	450	5.0
150	2.5*	460	5.0*
160	2.5	470	5.0
170	2.5	480	5.0
180	2.5*	490	5.0
190	2.5	500	6.0*
200	3.0**	510	6.0
210	3.0	520	6.0
220	3.0*	530	6.0
230	3.0	540	6.0*
240	3.0	550	6.0
250	3.0	560	6.0
260	3.0**	570	6.0
270	3.0	580	6.0**
280	3.0	590	8.0
290	3.0	600	8.0**
300	4.0*	610	8.0
310	4.0	620	8.0*
320	4.0	630	8.0
330	4.0	640	8.0
340	4.0*	650	8.0*
350	4.0		

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4.3.2 POINTER STOP ADJUSTMENT.- The maximum-speed pointer stop shall be so adjusted that the maximum-speed pointer will not stop until 650 knots is reached.

4.3.2.1 AIRSPEED DRUM.- When tests on the airspeed pointer are specified, the tests shall include the airspeed drum.

4.4 TEST METHODS

4.4.1 INDIVIDUAL TESTS

4.4.1.1 LEAKAGE

4.4.1.1.1 CASE LEAKAGE.- The indicator shall not be subjected to the 10 inches of mercury pressure.

4.4.1.1.2 DIAPHRAGM CAPSULE LEAK.- The diaphragm capsule leak test shall be accomplished only on the airspeed pointer.

4.4.1.2 MAXIMUM SPEED SCALE ERROR AT ROOM TEMPERATURE.- The maximum speed pointer shall be tested for scale errors at the points of the scale indicated by asterisks in table IV. The test shall be accomplished by adjusting the Mach number adjustment to the setting listed at the top of table IV. During this test, the pitot and static connection shall be connected together to a standard mercury barometer and to sources of vacuum and pressure. The reduction in pressure, as indicated by the barometer, shall be made at a rate corresponding to an increase in altitude of approximately 6,000 fpm. The scale errors shall not exceed the tolerances specified in table IV. This test shall be combined with the hysteresis test at each Mach number setting.

4.4.1.3 HYSTERESIS.- Not more than 15 minutes after the indicator has been subjected to the pressure corresponding to the highest altitude specified in table IV, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 3,000 fpm until the pressure corresponding to 30,000 feet is reached. The indicator shall remain at this pressure for at least 5 minutes, but not more than 15 minutes, before the test reading is taken. After the reading has been taken, the pressure shall be further increased at the above rate until the pressure corresponding to 10,000 feet is reached. The indicator shall remain at this pressure for at least 1 minute, but not more than 10 minutes, before a test reading is taken. After the reading has been taken, the pressure shall be further increased at the above rate until atmospheric pressure is reached. The reading of the maximum speed pointer at either of the two test points shall not differ from the corresponding reading, with decreasing pressure, by more than 2 knots.

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TABLE IV

MACH NUMBER										
	0.6		0.7		0.8		0.9		1.0	
Altitude 1,000 Feet	Airspeed Knots	Tolerance Knots	Airspeed Knots	Tolerance Knots	Airspeed Knots	Tolerance Knots	Airspeed Knots	Tolerance Knots	Airspeed Knots	Tolerance Knots
0	397**	4	463*	6	529*	8	595*	10	--	--
5	364	4	426	5	488	6	550	8	613	10
10	333*	4	390*	4	448*	6	507*	6	566*	8
15	303	4	356	4	410	5	464	6	520	8
20	275**	4	323*	4	373*	4	423*	5	475*	6
25	248	4	292	4	337	4	384	4	431	5
30	223*	4	262*	4	303*	4	346*	4	390**	4
35	199	4	235	4	272	4	310	4	350	4
40	177**	4	209*	4	242*	4	277*	4	312*	4
45	157	4	186	4	216	4	247	4	279	4
50	140*	4	165*	4	192*	4	220*	4	248**	4

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4.4.1.4 AFTER EFFECT.- Not less than 1 minute, and not more than 5 minutes, after the completion of the maximum speed scale at room temperature and hysteresis tests, the maximum speed pointer shall be checked to determine that it has returned to its original reading, corrected for any change in atmospheric pressure, within 4 knots.

4.4.1.5 FRICTION MAXIMUM SPEED POINTER.- The indicator shall be tested for friction at every other one of the points marked with an asterisk in table IV beginning with the first asterisk. The pressure shall be decreased to bring the pointer to the desired reading and held constant while the two readings are taken, the first before the indicator is tapped and the second after the indicator is tapped. The difference between any two readings shall not exceed the tolerance specified in table II. The pointer shall move smoothly while the pressure is varied uniformly without vibration of the indicator. This test may be conbined with the scale error test.

4.4.1.6 POSITION ERROR INDICATED AIRSPEED POINTER AND MAXIMUM SPEED POINTER.- The pointer reading, taken while the indicator is held in any desired position, and while it is being tapped, shall not differ from its reading, when held in normal operating position, by more than the amount specified in table II. At least one of these positions shall be at approximately 45° with the horizontal, simulating a 45° climb or dive or both. This test shall be conducted at the test points marked with a double asterisk in tables III and IV. This test may be combined with the friction scale error test.

4.4.1.7 INTEGRAL LIGHTING.- The lighting circuit shall be properly connected and tested to show compliance with the requirements specified in 3.9 and 3.9.1.

4.4.2 SAMPLING PLANS AND TESTS

4.4.2.1 SAMPLING PLAN A.- One indicator shall be selected at random from each 100 or less produced and subjected to the following tests:

4.4.2.1.1 DRUM-POINTER RELATIONSHIP.- The indicated airspeed pointer shall be set on the graduation and the drum indication shall be noted. The difference between the respective readings of the drum and pointer shall not differ by more than 2 knots. This test shall be conducted at the points marked with a double asterisk in table III.

4.4.2.1.2 LOW TEMPERATURE OPERATION.- The test points shall be every other point in tables III and IV marked with an asterisk, beginning with the first asterisk. The scale errors in this test shall not exceed the tolerances specified in tables III and IV by more than 2 knots, but in no case shall the total allowable scale error exceed 10 knots. During

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and after this test, the maximum airspeed pointer stop adjustment shall operate properly. The hysteresis and after effect tests shall not be repeated at low temperature.

4.4.2.1.3 HIGH TEMPERATURE OPERATION.- The test points shall be every other point in tables III and IV marked with an asterisk, beginning with the first asterisk. The scale errors in this test shall not exceed the tolerance specified in tables III and IV by more than 2 knots, but in no case shall the total allowable scale error exceed 10 knots. During and after this test, the maximum airspeed pointer stop adjustment shall operate satisfactorily. The hysteresis and after effect tests shall not be repeated at high temperature.

4.4.2.1.4 VIBRATION ERROR.- The Mach Number setting shall be that specified in table II and the maximum amplitude of pointer vibrations and variations, from their original positions, shall not exceed the tolerance specified in table II.

4.4.2.1.5 MAXIMUM POINTER STOP.- The indicator shall be disconnected from external sources of vacuum or pressure and, adjusted to 1.0 Mach number, the stop mechanism tested for range by turning the screw in the lower left mounting lug to determine the limits of adjustment. The adjustment shall operate smoothly throughout the range of 650 to at least 300 knots on the airspeed scale. No damage or loosening in the mechanism shall result from this test.

4.4.2.2 SAMPLING PLAN B

4.4.2.2.1 LOW TEMPERATURE EXPOSURE.- The scale errors shall not exceed the tolerance specified for the low temperature test.

4.4.2.2.2 VIBRATION FAILURE.- The Mach number setting, the differential pressure, and altitude shall be as specified in table II. The difference between the indication of the pointers at the beginning and at the end of the 3-hour vibration period shall not exceed the tolerance specified in table II.

4.5 REJECTION AND RETEST.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.5.1 INDIVIDUAL TESTS MAY CONTINUE.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

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4.5.2 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

5. PREPARATION FOR DELIVERY

5.1 PACKAGING

5.1.1 LEVEL A.- When level A packaging is specified (see 6.2), each indicator shall be individually packaged in accordance with MIL-P-7936 for level A. The method of preservation shall conform to MIL-P-116, method II.

5.1.2 LEVEL C.- When level C packaging is specified (see 6.2), each indicator shall be individually packaged in accordance with standard commercial practice.

5.2 PACKING

5.2.1 LEVELS A AND B.- When levels A and B are specified (see 6.2), the indicator shall be packed in accordance with MIL-P-7936 for levels A and B.

5.2.2 LEVEL C.- When level C is specified (see 6.2), the indicators shall be packed in accordance with standard commercial practice.

5.3 MARKING OF SHIPMENTS.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Indicator, Indicated Airspeed AVU-10/A,
Integrally Lighted

5.3.1 REINSPECTION DATE.- The reinspection date markings shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 INTENDED USE.- The AVU-10/A airspeed indicator covered by this specification is intended to indicate the airplane's speed and the maximum allowable speed as set by the Mach number adjustment at the rear of the case. This is accomplished by the use of 2 pointers.

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6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging and packing.

6.3 QUALIFICATION.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Wright Air Development Division, Wright-Patterson Air Force Base, Ohio and information pertaining to qualification of products may be obtained from that activity.

Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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